3.4.2. *The case of measures.* The above insight, which consists in reweighing the metric with respect to which the gradient is being taken, can be formally adapted to Wasserstein setting for (3.4)—which we recall, is not a gradient flow for the standard Wasserstein gradient of E_β—. But note that (3.4) writes as

$$\partial_t \mu(t) + \operatorname{div}\left(\frac{\nabla \delta \mathsf{E}_\beta[\mu(t)]}{\delta \mathsf{E}_\beta[\mu(t)]}\mu(t)\right) = 0, \tag{3.15}$$

with $\delta \mathsf{E}_\beta[\mu](x) = \int e^{\beta\langle x,y\rangle}\,\mathrm{d}\mu(y)$. To avoid technicalities we henceforth only focus on the absolutely continuous case. For a fixed $\mu \in \mathcal{P}_{\mathrm{ac}}(\mathbb{S}^{d-1})$, as in the well-known formal Riemannian reinterpretation of the Wasserstein space using Otto calculus [Ott01], [CNWR24, Chapter 5], we consider

$$\mathrm{T}_\mu \mathcal{P}_{\mathrm{ac}}(\mathbb{S}^{d-1}) = \overline{\{\nabla\psi \colon \psi \in C^\infty(\mathbb{S}^{d-1})\}}^{L^2(\mu)},$$

which, rather than endowing with the standard formal metric tensor given by the $\dot{H}^1$–inner product

$$\langle \nabla\psi_1, \nabla\psi_2\rangle_\mu := \int \langle \nabla\psi_1(x), \nabla\psi_2(x)\rangle\,\mathrm{d}\mu(x),$$

we instead endow with

$$\langle \nabla\psi_1, \nabla\psi_2\rangle_{\mu,\mathsf{E}_\beta} := \int \langle \nabla\psi_1(x), \nabla\psi_2(x)\rangle\,\delta\mathsf{E}_\beta[\mu](x)\,\mathrm{d}\mu(x).$$

Continuing the Riemannian geometry analogy, through this metric tensor we can define a distance between $\mu_0, \mu_1 \in \mathcal{P}_{\mathrm{ac}}(\mathbb{S}^{d-1})$ by solving the variational problem

$$\inf_{(\mu(t),v(t))_{t\in[0,1]}} \left\{ \int_0^1 \|v(t)\|^2_{\mu(t),\mathsf{E}_\beta}\,\mathrm{d}t \colon (\mu, v) \text{ satisfy (3.16)}, \mu(0) = \mu_0, \mu(1) = \mu_1 \right\},$$

where

$$\partial_t \mu(t,x) + \operatorname{div}(v(t,x)\mu(t,x)) = 0 \qquad \text{on } [0,1]\times\mathbb{S}^{d-1}. \tag{3.16}$$

This variational problem is a generalization of the celebrated Benamou-Brenier formula [BB00], the value of which we dub $W^2_{2,\mathsf{E}_\beta}(\mu_0,\mu_1)$: it is a weighed Wasserstein distance. For a curve of measures $(\mu(t))_{t\geqslant 0}$ with tangent vectors $(v(t))_{t\geqslant 0}$ (meaning they solve (3.16)), the Wasserstein gradient $\mathbb{W}_{\mathsf{E}_\beta}\mathsf{E}_\beta$ induced by this geometric setup is then the element of $\mathrm{T}_{\mu(t)}\mathcal{P}_{\mathrm{ac}}(\mathbb{S}^{d-1})$ such that

$$\partial_t \mathsf{E}_\beta[\mu(t)] = \langle \mathbb{W}_{\mathsf{E}_\beta}\mathsf{E}_\beta[\mu(t)], v(t)\rangle_{\mu(t),\mathsf{E}_\beta}.$$

We can now demonstrate that the vector field driving (3.15) is the Wasserstein gradient of E_β corresponding to this geometric setup. Indeed as in [CNWR24, Definition 5.9] we first have

$$\partial_t \mathsf{E}_\beta[\mu(t)] = \int \delta\mathsf{E}_\beta[\mu(t)]\,\mathrm{d}\partial_t\mu(t).$$

We then find

$$\int \delta\mathsf{E}_\beta[\mu(t)]\,\mathrm{d}\partial_t\mu(t) = \int \langle \nabla\delta\mathsf{E}_\beta[\mu(t)], v(t)\rangle\,\mathrm{d}\mu(t) = \left\langle \frac{\nabla\delta\mathsf{E}_\beta[\mu(t)]}{\delta\mathsf{E}_\beta[\mu(t)]}, v(t)\right\rangle_{\mu(t),\mathsf{E}_\beta},$$

as desired. The literature studying weighed Wasserstein distances such as the one above is rather scarce, but a relevant reference is [Li21].

Part 2. Clustering

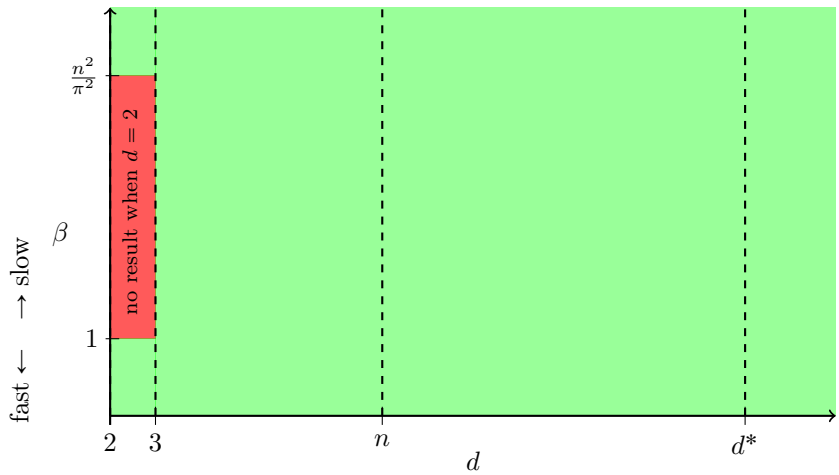

Figure 2. Green zones indicate regimes where convergence to a single cluster as $t \to +\infty$ can be proven. Here $n \geqslant 2$ is fixed. When d is larger than specific thresholds, the long-time asymptotics can be chiseled out in finer detail. Convergence is slow when $\beta \gg 1$ (relative to the size of d, n), as even the exponential decay constant when $d \geqslant n$ is of the form $\lambda = O(e^{-\beta})$ and thus degenerates. One rather expects dynamic metastability. Section 4 addresses the case where β is small and Section 5 where it's large, whereas Section 6 covers the high-dimensional case at arbitrary β.

As alluded to in the introductory discussion, clustering is of particular relevance in tasks such as sentiment analysis, masked language modeling, summarization, and so on. Therein, the output measure encodes the probability distribution of the missing tokens for instance, and its clustering indicates a small number of possible outcomes. In Sections 4, 5, and 6, we show several results (mostly summarized in Figure 2) which indicate that the limiting distribution is a point mass. While it may appear that this leaves no room for diversity or randomness, which is at odds with practical observations, these results hold for the specific choice of parameter matrices, and apply in possibly very long-time horizons. Numerical experiments indicate a more subtle picture for different parameters—for instance, there is an appearance of a long metastable phase during which the particles coalesce in a small number of clusters, which seems consistent with behavior in trained models (Figure 1 in Section 6.3)—. We are not able to theoretically explain this behavior as of now.

Ultimately, the appearance of clusters is somewhat natural since the Transformer dynamics are a weighted average of all particles, with the weights being hard-wired to perform a fast selection of particles most similar to the i-th particle being queried. This causes the emergence of leaders which attract all particles in their vicinity. In the natural language processing interpretation, where particles represent tokens, this further elucidates the wording *attention* as the mechanism of inter-token

attraction, and the amplitude of the inner product between tokens can be seen as a measure of their *semantic similarity*.

4. A single cluster for small β

As seen in Figure 2, the dimension d and inverse temperature β appear to play a key role in the clustering results we obtain. In this section and in Section 5, we begin by focusing on extreme choices of β whilst d, n are fixed. We first focus on the case $\beta = 0$ (Section 4.1), before moving to the case $\beta \ll 1$ by a perturbation argument (Section 4.2). We cover the case where β is sufficiently large, but finite, in Section 5. The case $\beta = +\infty$ is of little interest since all particles are fixed by the evolution.

4.1. **The case $\beta = 0$.** For $\beta = 0$, both (SA) and (USA) read as

$$\dot{x}_i(t) = \mathbf{P}^{\perp}_{x_i(t)} \left(\frac{1}{n} \sum_{j=1}^{n} x_j(t) \right), \qquad t \geqslant 0. \tag{4.1}$$

The following result shows that generically over the initial points, a single cluster emerges. It complements a known convergence result ([FL19, Theorem 2]) for (4.1). In [FL19, Theorem 2], the authors show convergence to an antipodal configuration, in the sense that $n-1$ particles converge to some $x^* \in \mathbb{S}^{d-1}$, with the last particle converging to $-x^*$. Moreover, once convergence is shown to hold, it holds with an exponential rate. Mimicking the proof strategy of [BCM15, Theorem 2.2] and [HKR18, Theorem 3.2], we sharpen this result by showing that the appearance of an antipodal particle is non-generic over the choice of initial conditions.

Theorem 4.1. *Let $d, n \geqslant 2$. For Lebesgue almost any initial sequence $(x_i(0))_{i\in[n]} \in (\mathbb{S}^{d-1})^n$, there exists some point $x^* \in \mathbb{S}^{d-1}$ such that the unique solution $(x_i(\cdot))_{i\in[n]} \in C^0(\mathbb{R}_{\geqslant 0}; (\mathbb{S}^{d-1})^n)$ to the corresponding Cauchy problem for (4.1) satisfies*

$$\lim_{t\to+\infty} x_i(t) = x^*$$

for any $i \in [n]$.

This is also referred to as convergence toward *consensus* in collective behavior models. We refer the interested reader to Appendix A for the proof, which relies on a gradient flow reinterpretation of (4.1), much like the proofs of several subsequent theorems. We provide some comments in Section 5.

4.2. **The case $\beta \ll 1$.** Theorem 4.1 has some implications for small but positive β, something which is already seen in Figure 3 and Figure 4. This is essentially due to the fact that, formally,

$$\dot{x}_i(t) = \mathbf{P}^{\perp}_{x_i(t)} \left(\frac{1}{n} \sum_{j=1}^{n} x_j(t) \right) + O(\beta)$$

for $\beta \ll 1$. So, during a time $\ll \beta^{-1}$, the particles do not feel the influence of the remainder $O(\beta)$ and behave as in the regime $\beta = 0$. This motivates

Theorem 4.2. *Fix $d, n \geqslant 2$. For $\beta \geqslant 0$, let $\mathcal{S}_\beta \subset (\mathbb{S}^{d-1})^n$ be the subset consisting of all initial sequences for which the associated solution to the Cauchy problem for (SA) (or (USA)) converges to one cluster as $t \to +\infty$. Then*

$$\lim_{\beta\to 0} \mathbb{P}(\mathcal{S}_\beta) = 1.$$